

Business Insights & Recommendations

E-Commerce Delivery Performance Analysis | Olist Brazilian E-Commerce Public Dataset

1. Executive Summary

Among 96,470 eligible delivered orders, 6.77% were delayed and 93.23% were delivered on time. Delayed orders were, on average, 10.62 days late. SP and RJ had the highest delayed-order volumes, together accounting for approximately 50.7% of all delayed orders. Delayed orders were also associated with substantially lower average review scores: 2.27/5 versus 4.29/5 for on-time orders.

2. Key Findings

Finding	Evidence	Business Meaning
Delivery delays are measurable	6,534 delayed orders; 6.77% delay rate.	Delivery performance is generally strong, but a meaningful group of orders experiences delays.
Delays are geographically concentrated	SP: 1,820; RJ: 1,495; ~50.7% combined.	A small number of high-volume states account for a large share of delays.
Delay severity varies by state	Some states have high average delays but very small delayed-order volumes.	Average delay should not be used alone to set priorities.
Delayed orders have lower observed satisfaction	2.27/5 delayed vs 4.29/5 on time; difference 2.02 points.	Delivery performance should be monitored alongside customer experience.

3. Business Insights

3.1 Overall Delivery Performance

Of 96,470 eligible delivered orders, 6,534 were delayed and 89,936 were on time. This gives a 6.77% delay rate and 93.23% on-time rate. When delays occurred, the average delay was 10.62 days.

3.2 Geographic Concentration

SP recorded 1,820 delayed orders and RJ recorded 1,495. Together they represented approximately 50.7% of all delayed orders, making them practical starting points for operational investigation.

3.3 Delay Severity and Volume

Some states show very high average delay values, but these can be based on only a few delayed orders. Therefore, average delay should be interpreted together with delayed-order volume and delay rate.

3.4 Customer Satisfaction

Delayed orders had an average review score of 2.27/5 compared with 4.29/5 for on-time orders. This is a 2.02-point difference. The analysis indicates an association, not causation.

4. Recommendations

4.1 Prioritise SP and RJ for investigation

Begin operational investigation in SP and RJ because they contribute the largest volumes of delayed orders.

4.2 Use a combined prioritisation approach

Evaluate states using delayed-order volume, delay rate, and average delay rather than relying on one metric.

4.3 Monitor customer satisfaction with delivery KPIs

Track review scores for delayed and on-time orders alongside delivery performance to identify customer-impact signals.

4.4 Use the Power BI dashboard for operational visibility

Use the dashboard as a concise monitoring view for delivery KPIs, state-level delay volume, and customer-rating differences.

5. Analytical Considerations

- Only delivered orders with a valid customer delivery date are included in delivery-delay calculations.
- Orders without an actual delivery date are excluded from delivery-time and delay calculations.
- Where an order has multiple review records, review scores are averaged at order level.
- The analysis identifies association, not causation, between delivery delays and customer satisfaction.
- High average delay values based on very small delayed-order volumes should be interpreted cautiously.

6. Conclusion

The analysis shows that delivery performance is generally strong, but delays are concentrated in specific geographic areas and are associated with substantially lower customer review scores. SP and RJ are the most practical starting points for further investigation. A balanced view of delay volume, delay rate, delay severity, and customer satisfaction provides a stronger basis for prioritisation.